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1) Lead (Pb)

a) Highly toxic heavy metal that accumulates in bones, kidneys and the brain.

b) Common Sources: batteries, paints, pipes, e-waste and industrial emissions.

c) Frequently detected in soil, dust, Sediment and landfill leacheate near industrial areas.

2) Chromium (Cr)

a) Exists mainly as Cr(III) (essential trace nutrient) and Cr(VI) (highly toxic and carcinogenic)

b) Common Sources: electroplating, leather tanning, Stainless steel, textile industries.

c) Cr(VI) is much more mobile & toxic in water than Cr(III)

3) Arsenic (As)

a) Naturally occurs in rocks and groundwater; also released from mining and pesticides.

b) Long-term exposure causes skin lesions, cancers, cardiovascular diseases and diabetes.

4) Cadmium (Cd)

a) found in rechargeable batteries, pigments electroplating and e-waste.

b) Accumulates in kidneys and has a very long biological half-life.

5) Mercury (Hg)

a) Exists as elemental, inorganic, and organic forms; methylmercury is the most toxic

b) Common Sources: fluorescent lamps, thermometers, batteries, coal combustion and e-waste.